Akka vs. Vercel AI SDK

A comparison for teams building agentic AI

October 2026

The Vercel AI SDK gives you outstanding developer experience for building and shipping AI features — but you own production. It is a TypeScript developer SDK with serverless hosting and the Vercel Workflows durability service: no active-active HA/DR with zero-byte RPO, no six-nines SLA on the running agent, and no governance beyond AI SDK agents and AI Gateway traffic. The Akka Agentic AI Platform owns the running system end to end and guarantees resilience and scalability.

99.99%	99.9999%	4ms	5B
VERCEL ENTERPRISE SLA	AKKA SLA	AKKA STATE READS	TOKENS / SEC BENCHMARK

DIMENSION	VERCEL AI SDK	AKKA
What it is	A TypeScript developer SDK for building AI apps and agents, with serverless hosting (AI Cloud / Fluid Compute)	The Akka Agentic AI Platform, which guarantees resilience and scalability, with an enterprise durable runtime
Scope	Model calls, tool loops, streaming, UI hooks, MCP — outstanding DX for building; you own production	Orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime
Language	TypeScript / JavaScript only (React, Next.js, Vue, Svelte, Node.js)	Java / Scala; deploy on Akka cloud, hyperscaler VPC, own Kubernetes, on-prem, sovereign cloud
Availability SLA	99.99% on Enterprise, commercially reasonable efforts, excludes Excused Downtime	99.9999% — entire platform, backed by indemnities
HA / DR	Serverless redundancy; no published active-active SLA, RPO, or RTO commitment	Active-active HA/DR; sub-1-minute RTO; zero-byte RPO
Durable state	Stateless serverless functions; durability via event-log replay in Vercel Workflows, one home region per run	Durable in-memory, 4ms reads / sub-10ms writes, replayable from its event journal
Governance / EU AI Act	Security/privacy certifications; tool approvals (AI SDK 7) and AI Gateway model rules; no named AI regulation	Aspect-woven runtime enforcement + full pre-production governance
Cost model	Consumption-metered compute (Active CPU, memory, invocations) + Workflows events and storage	Shared compute; up to 90% lower infrastructure for the same workload
Certifications	SOC 2 Type II, ISO 27001:2022, PCI DSS, HIPAA, GDPR, EU-U.S. DPF, TISAX AL2 (reports on request)	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, EU AI Act, NIST AI RMF)

The Vercel AI SDK Is a Developer SDK with Hosting; Akka Is an Enterprise Platform

The Vercel AI SDK is the TypeScript toolkit for building AI-powered applications and agents — and it is excellent at that. It abstracts away the differences between model providers behind one clean API, and AI SDK 6 added first-class agents, loop control, human-in-the-loop tool approval, full MCP support, and DevTools. AI SDK 7 (June 2026) adds HarnessAgent for Claude Code, Codex, and Pi, the durable WorkflowAgent, and signed tool approvals. Paired with Vercel's serverless hosting (AI Cloud and Fluid Compute), a team can build and ship an AI feature with remarkable speed.

That speed is real, and it is the right tool for building. The line is **production ownership**. The SDK is a library that runs on Vercel Functions or any other Node.js host; durability comes from Vercel Workflows, a separately metered product, and the SDK carries no availability guarantee for the running agent. With the Vercel AI SDK you assemble and operate the failover, the availability, and the governance around the agent. With Akka they are the platform.

Everything an agentic system needs runs in one Akka runtime, so a system that compiles is ready to run in production. The Akka SDK models the entire agentic system in one place — agents, tools, durable memory, streaming, APIs, and governance — and the platform generates it, tests it, and runs it as one certified service.

Capability	Vercel AI SDK	Akka
Build AI features / agents in code	Yes — best-in-class DX	Yes
Model abstraction, tool loops, streaming, MCP	Yes	Yes
Native durable runtime for the running agent	Vercel Workflows, a separate metered product	Built in
Active-active HA/DR (zero-byte RPO)	Not published	Built in
Six-nines availability SLA on the agent	99.99% Enterprise	99.9999%
Durable in-memory state	Managed persistence, replayed; one home region per run	Built in, 4ms / sub-10ms
Embedded runtime governance	Tool approvals in agent code; AI Gateway model rules	Inline, runtime-embedded
Pre-production governance	Not in the SDK	Classification, sign-offs, sealed posture

Availability, Disaster Recovery, and Durable State

Vercel publishes a 99.99% availability SLA on its Enterprise plan, made on a commercially reasonable efforts basis and excluding Excused Downtime — outages caused by third-party vendors, the internet, your own software, or factors outside Vercel's reasonable control. Service credits are capped at 50% of the monthly fee, and the customer must file a claim with log files. There is no published active-active multi-region SLA, no zero-byte RPO, and no RTO commitment on the running agent.

Metric	Vercel AI SDK	Akka
Availability SLA	99.99% (Enterprise, commercially reasonable efforts)	99.9999%
Allowed downtime / year	~52 minutes	~31 seconds
RTO	Not committed	Sub-1 minute
RPO	Not committed	Zero byte
SLA scope	Hosting platform; Excused Downtime excluded	The entire running platform

State follows the same line. Vercel's compute is serverless — Fluid Compute functions are short-lived and reused, with a maximum duration of 800 seconds (1800s in beta); a function that exceeds it returns a 504 timeout. Durability for long-running agents comes from Vercel Workflows, built on the open-source Workflow SDK and generally available since April 2026, which records each step to an event log and replays it, with each run's state kept in one home region. Akka holds state in **durable sharded in-memory** at 4ms reads and sub-10ms writes, replayable from its own event journal — no external store on the hot path.

Up to 90% Cheaper to Operate

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required to run the same agentic transaction volume, not list price. The drivers are actor concurrency (~10T tokens/core/year vs ~2T; ~80% less compute than Python frameworks), shared compute across orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime, and micro-checkpointing that minimizes retries. Manulife reported up to 300% more concurrency and 30–50% faster processing after porting from Python.

Vercel's serverless model bills consumption: Active CPU while code executes, provisioned memory for running instances, and invocations, with Vercel Workflows billed per event and per GB written and retained. Spend moves with load, and the meter covers the hosting compute — the agent runtime, memory, streaming, and governance you assemble around it carry their own cost. Akka runs all of it on one shared-compute runtime for a fixed annual fee finance can forecast.

The Language Boundary

The Vercel AI SDK is **TypeScript / JavaScript only** — React, Next.js, Vue, Svelte, Node.js. That is the right surface for the web tier and for building AI features quickly. It also means an enterprise standardizing on it commits its agentic systems to the JS/TS runtime.

Akka runs on the JVM (Java / Scala) and deploys anywhere — Akka cloud, hyperscaler VPC (AWS/Azure/GCP), the customer's own Kubernetes, on-prem, or sovereign cloud with country-isolated data and local SREs. Specs are portable across those targets, so the deployment model is a procurement choice, not a constraint.

Continuous AI Intelligence

Akka Optimize keeps improving the AI after it ships. It runs an evaluation loop on your own production traffic, grading live interactions and training smaller specialized models on your proprietary data, so the system keeps getting better as it runs. The loop delivers three results that compound over time:

- **Model leverage** — route each request to the best model from any vendor under your policies, and reserve frontier models for the requests that need them.
- **Data sovereignty** — the specialized models are trained on your proprietary data, owned by you, and run inside your own environment.
- **Cost governance** — full visibility and control of AI spend, with savings that compound as the loop keeps running.

The evaluations run continuously inside Akka and grade traffic from agents wherever they run, in Akka or in third-party harnesses.

Governance and the EU AI Act

Vercel holds security and privacy certifications — SOC 2 Type II, ISO 27001:2022, PCI DSS, HIPAA, GDPR, the EU-U.S. Data Privacy Framework, and TISAX AL2 — covering the hosting platform. AI SDK 7 holds a tool call for human approval, with optional HMAC-signed approvals, and AI Gateway denies requests for unapproved models and providers. Engineers write those approvals into each agent's TypeScript code, and the controls cover AI SDK agents and model calls routed through AI Gateway. AI Gateway logs each request's model, tokens, and cost and deletes prompts and responses after each request by default. Vercel maps its controls to no AI regulation such as the EU AI Act.

The penalties are enforceable now

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover

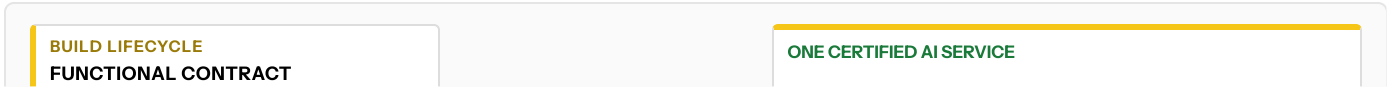
High-risk AI carries a 10-year logging-retention obligation (Art. 72).

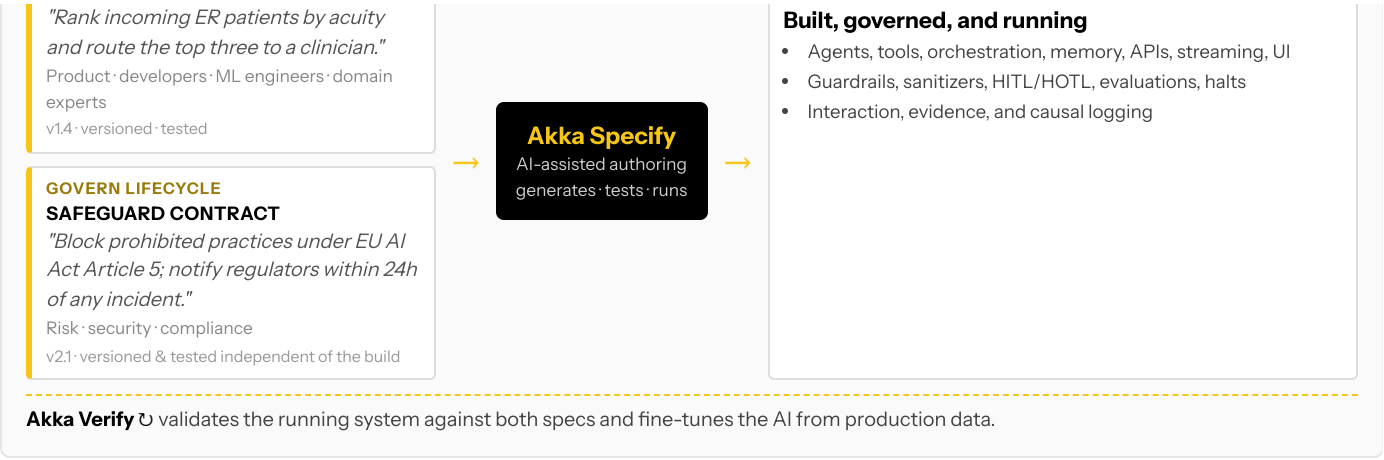
How Akka governs

At the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL control; classification against 241 regulations and frameworks and 1,718 controls before a system ships; multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. Governance a Vercel AI SDK team writes into each agent, Akka Verify applies to any AI system.

Two Lifecycles, One Certified System

Building with the Vercel AI SDK means TypeScript engineers writing agent code; there is no built-in path for a product manager, domain expert, or risk officer to author and version a governance contract independently. Akka runs two independent lifecycles on one platform via **Akka Specify**:





The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences — a workflow the Vercel AI SDK has no equivalent for.

Real-Time Streaming at Petabyte Scale

The Vercel AI SDK streams model tokens to the client, which is exactly what an interactive AI feature needs. It is not a streaming data engine. Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

For the Buyer: Risk, Compliance, and Accountability

Buyer concern	Vercel AI SDK	Akka
Certifications & audits	SOC 2 Type II, ISO 27001:2022, PCI DSS, HIPAA, GDPR, EU-U.S. DPF, TISAX AL2 (reports on request)	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Scope of accountability	The hosting platform; you build, integrate, and operate the agent runtime, memory, and governance	One platform, one SLA, 24/7 SRE — Akka owns the running system
Risk transfer	Standard cloud terms; SLA credits capped at 50% of monthly fee	Availability and data-integrity guarantees backed by contractual indemnities
Track record & funding model	Venture-funded: $9.3B Series F (Sept 2025), ~$863M raised; ~$340M ARR run-rate (Mar 2026); profitability not disclosed	Profitable and self-funding; since 2009 and 100,000+ deployments (52 banks); Dell Technologies Capital is largest shareholder, a customer, and an AI partner
Budget predictability	Consumption-metered serverless compute that scales with load	Fixed annual fee finance can forecast

Vercel is well-capitalized and growing fast, and its developer experience is genuinely best-in-class. The decision is scope and accountability: the Vercel AI SDK gives developers an outstanding way to build and ship AI features; Akka gives you the enterprise platform that owns the running system — durable runtime, six-nines availability, active-active HA/DR, and embedded governance.

Customers Running Agentic and Real-Time Systems on Akka

Manulife
2,000
developers across 100 projects on one governed platform

Tubi
5B tok/s
real-time hyper-personalization engine

Swiggy
71ms
order-assignment AI, ~50% latency reduction

John Deere
1,000+
tractor sensors turned into real-time insight

Verizon
750%
order-processing capacity gain; 6s → 2.4s response

Common Questions

We already use the Vercel AI SDK and love the developer experience. Why add Akka?
The Vercel AI SDK is an excellent way to build and ship AI features — keep using it for that. The question is what owns production: cross-

Can we add governance on top of the Vercel AI SDK?
AI SDK 7 adds human approval of tool calls and AI Gateway adds model and provider rules, written by the engineers who build each agent. Those controls cover AI SDK agents and AI Gateway traffic and

region failover, the availability guarantee on the running agent, and governance that reaches beyond each agent. Those are not in the SDK; you operate them yourself. Akka is the platform that owns the running system, with a 99.9999% SLA, active-active HA/DR, durable in-memory state, and embedded governance.

Don't Vercel Workflows and WorkflowAgent give me a durable runtime?

Vercel Workflows, built on the open-source Workflow SDK, adds durability to TypeScript and Python functions by recording each step to an event log and replaying it, with each run's state kept in one home region. That makes individual workflows crash-safe. It is not an active-active HA/DR runtime with a zero-byte RPO and a six-nines SLA on the running system. Akka provides durable sharded in-memory state at 4ms/sub-10ms and a contractual 99.9999% availability guarantee.

map to no named regulation. The EU AI Act expects immutable records witnessed as they happen and classification before a system ships. Akka embeds inline guardrails, hash-chained evidence, and pre-deployment classification against 241 regulations and frameworks and 1,718 controls, and proves conformance with Akka Verify.

Isn't the Vercel AI SDK cheaper because it's open source?

The SDK is free and open source, and that lowers the cost of building. Production cost is the infrastructure to run the workload: Vercel bills consumption-metered serverless compute that scales with load, and the agent runtime, memory, streaming, and governance around it carry their own cost. Akka's shared-compute model is up to 90% cheaper to operate for the same agentic transaction volume, on a fixed annual fee.

Sources

Vercel AI SDK (TypeScript scope): ai-sdk.dev/docs/introduction, github.com/vercel/ai — "the TypeScript toolkit"; React, Next.js, Vue, Svelte, Node.js; open-source library, runs on any host

AI SDK 7 (HarnessAgent, WorkflowAgent, signed tool approvals; June 25, 2026): vercel.com/blog/ai-sdk-7; npm ai 7.0.154 (Oct 2026)

AI SDK 6 (agents, loop control, MCP, DevTools, tool approval): vercel.com/blog/ai-sdk-6, ai-sdk.dev/docs/agents/overview, /agents/loop-control

Vercel AI Cloud / Fluid Compute: vercel.com/blog/the-ai-cloud-a-unified-platform-for-ai-workloads, vercel.com/docs/fluid-compute, /functions/limitations — serverless; 800s max (1800s beta); 504 timeout

Vercel Workflows / Workflow SDK / WorkflowAgent: vercel.com/docs/workflows, workflow-sdk.dev, Vercel Weekly 2026-04-20 — GA April 2026; event-log replay; one home region per run; Postgres World for self-hosting

Vercel Enterprise SLA: vercel.com/legal/sla, vercel.com/enterprise — 99.99% commercially reasonable efforts; Excused Downtime excluded; credits capped at 50% of monthly fee

Vercel certifications: vercel.com/docs/security/compliance, security.vercel.com — SOC 2 Type II, ISO 27001:2022, PCI DSS, HIPAA, GDPR, EU-U.S. DPF, TISAX AL2; reports on request

AI Gateway governance: vercel.com/docs/ai-gateway/security-and-compliance, vercel.com/changelog/ai-gateway-routing-rules — zero data retention by default, provider and model allowlists, Deny rules, budgets, request logs

Vercel funding & revenue: sacra.com/c/vercel, businesswire (Series F at $9.3B, Sept 2025), techcrunch.com (Apr 2026) — ~$863M raised; ~$340M ARR run-rate; profitability not disclosed

Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

Akka platform: 99.9999% availability, active-active HA/DR, sub-1 min RTO, zero byte RPO (contractual indemnities); durable in-memory 4ms / sub-10ms; 241 regulations and frameworks / 1,718 controls (699 with a financial penalty); 100,000+ deployments / since 2009; profitable; Dell Technologies Capital